

Freight Rate Prediction

Validation approach, data split, model selection, and the fixed-lane December forecast · Sogo Ogundowole

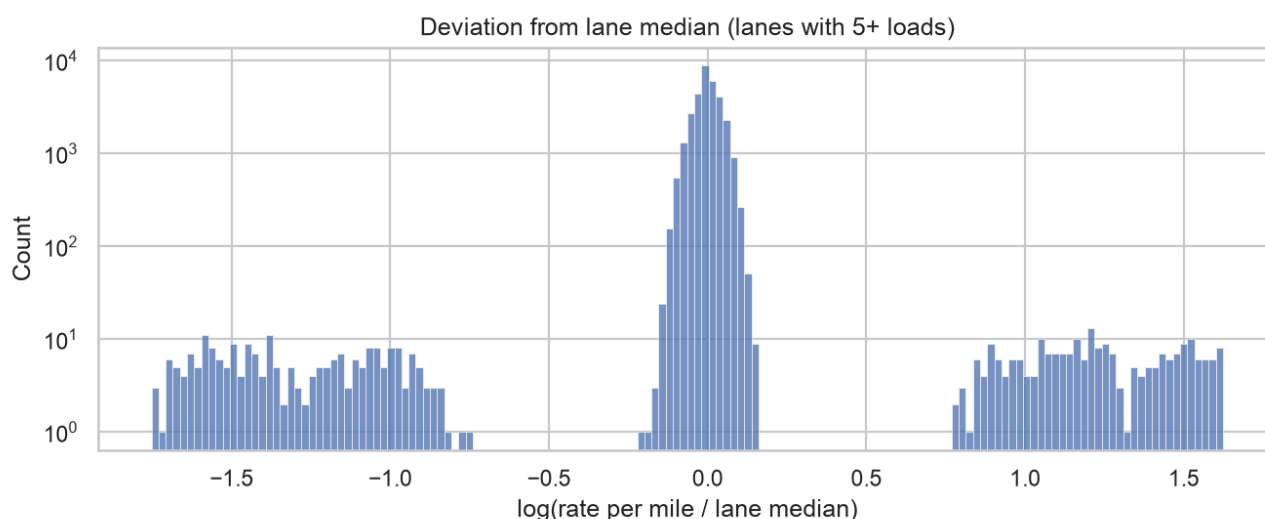
1. Summary

The task is to predict the posted rate for 12,000 truckload freight loads in November and December 2025, given ten months of labelled loads from January to October. The submitted model is a log-space blend of a Ridge regression and a LightGBM model, fitted on 47,323 loads after removing corrupted rates. On a two-month forward holdout it reaches a mean absolute error of \$48 and a mean absolute percentage error of 2.0% on loads with valid rates, and 4.5% when the corrupted loads are included. Every candidate was validated the same way, and the blend is the best model in each of three forward-chaining windows.

2. Data and quality issues

Each load carries origin and destination city with coordinates, distance, equipment type, weight, date, a market index, and a quote signal. The validation set covers the two months immediately after the training period, includes eight cities and about a third of lanes never seen in training, and carries the same feature defects as the training set. Four issues were identified and handled in `freight_rate.data`:

Issue	Evidence	Treatment
Negative weights (0.6% of loads)	Magnitudes mirror the valid range exactly and the rates on those loads match their lanes, so they are sign flips rather than different loads.	Absolute value.
Missing weight and market index (under 1% each)	Unrelated to the rate.	Equipment median and same-day mean respectively, with missing-indicator flags. The market index is a day-level series, so the same-day mean is a near-exact fill.
Corrupted rates (1.4% of loads)	Typical loads sit within a few percent of their lane median rate per mile; a separate population sits at three times or a quarter of it, with nothing in between.	Excluded from model fitting with a factor-of-two rule against the lane median, falling back to the equipment and distance-band median for thin lanes. Retained in evaluation.
Distance floored at 70 miles	High circuitry loads are all short lanes at the floor, and their rates match other short hauls.	No correction; circuitry kept as a feature.



Deviation of rate per mile from the lane median, log scale. The corrupted population is fully separated from ordinary variation.

3. Data split and validation approach

The loads to be predicted are strictly later than the training data, and rates drift through the year with the market index and an underlying trend. A random split would score the model on loads from the same days it trained on and overstate accuracy, so validation is time-ordered throughout.

1. **Primary holdout.** Fit on January to August (38,000 loads), score on September to October (9,500 loads). This is a two-month forward window with the same horizon as the November to December task, and it is the basis for the headline metrics.
2. **Forward-chaining folds.** Three windows, May-Jun, Jul-Aug, and Sep-Oct, each trained on all earlier months. These check that the ranking of models is stable across the year rather than specific to one window.
3. **Unseen geography.** Because a third of validation lanes are new, holdout error is reported separately for lanes seen and unseen in training. They are equal, at about 2% each, which confirms that the coordinate and distance features generalise.
4. **Metrics.** Mean absolute error, mean absolute percentage error, and R^2 on the raw dollar rate. Each is reported with the corrupted loads excluded, which isolates model differences, and with them included, which is the figure to expect against hidden labels that carry the same defect.

Cleaning statistics such as imputation medians are learned from the training portion of each split only. The outlier rule uses lane medians over the whole labelled set, but it only removes rows from the fitting portion and does not touch the target of any scored row.

4. Features

All models predict the logarithm of the rate, because the rate is multiplicative in its drivers: distance with a strongly non-linear rate per mile curve, a constant equipment multiplier, origin and destination effects of up to $\pm 10\%$, and a market index with an elasticity near 0.14. Features are: log distance and distance, great-circle distance and circuitry, equipment, origin and destination city, coordinates and their differences, weight, market index and its log, quote signal, day of week, and a linear time trend in days.

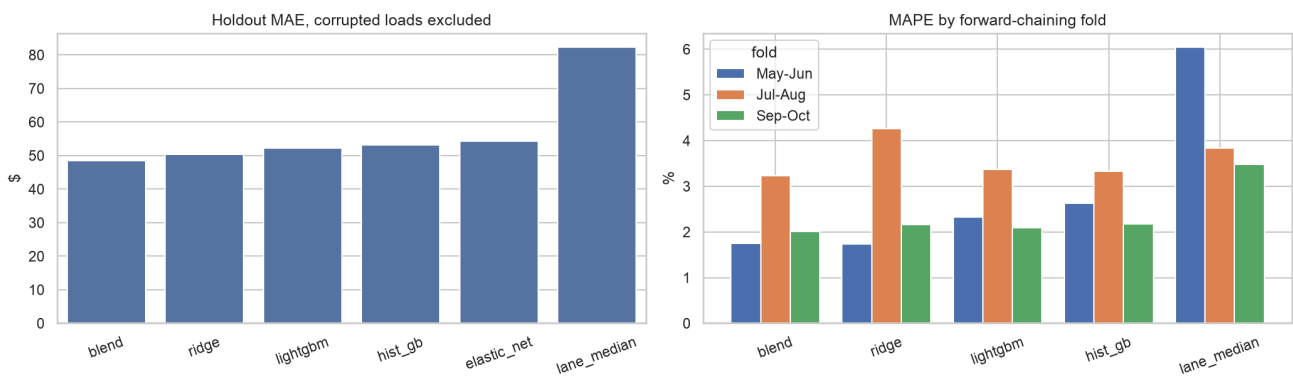
The trend was the most consequential feature. Rates rise about 0.5% per month independent of the market index. Without it, every model under-predicted the holdout by 4 to 5%, and adding it cut Ridge's

holdout error by more than half.

5. Model comparison

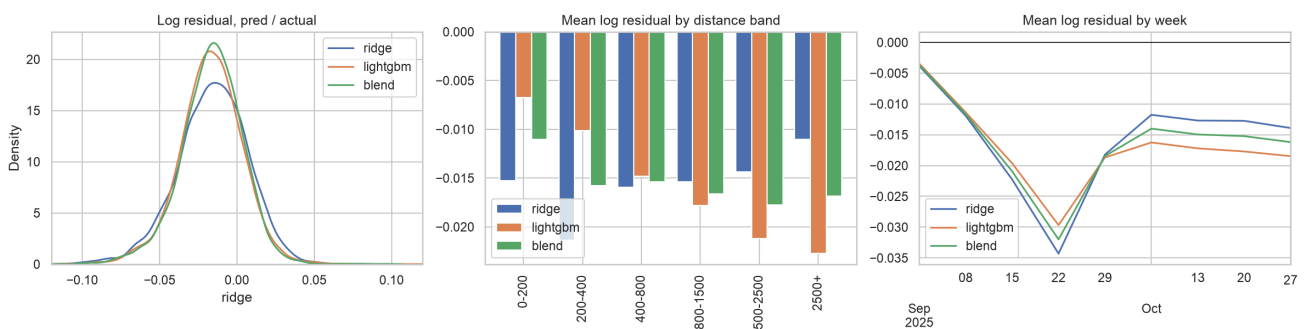
Models were introduced in order of complexity. Ridge uses a cubic spline in log distance and one-hot cities with unknown categories ignored, so a new city falls back to its coordinates. Gradient boosting uses ordinal-encoded categories with native categorical handling. The blend is the equal-weight geometric mean of the Ridge and LightGBM predictions.

Model	MAE, clean	MAPE, clean	R ² , clean	MAE, all	MAPE, all	Fold-mean MAPE
Lane-median baseline	82.4	3.48%	0.992	138.0	5.98%	4.46%
Ridge, log rate	50.3	2.16%	0.997	106.2	4.66%	2.72%
Elastic net, log rate	54.3	2.28%	0.996	110.1	4.77%	–
Histogram gradient boosting	53.2	2.17%	0.997	109.1	4.65%	2.71%
LightGBM	52.2	2.09%	0.997	108.0	4.58%	2.59%
Ridge + LightGBM blend (submitted)	48.4	2.01%	0.997	104.4	4.50%	2.33%



Holdout MAE (left) and MAPE by forward-chaining window (right).

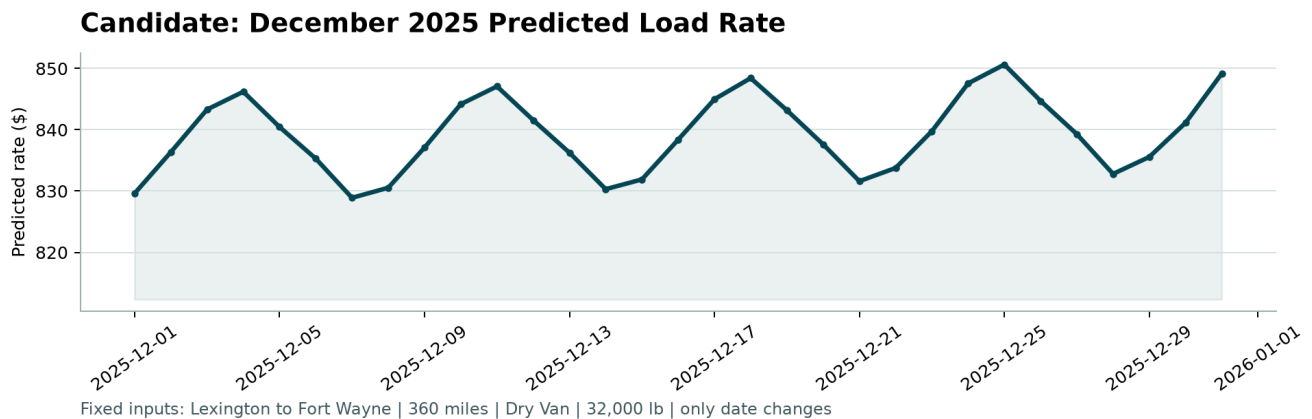
Why the blend. Ridge has the tightest residual spread of any single model; where it loses is level, in the one window (Jul-Aug) where the trend and the market index are nearly collinear in the training months. LightGBM handles the tails and the U-shaped quote-signal effect better but cannot extrapolate the trend past the last training day. Their errors are only partly correlated, so averaging them in log space beats both members in every window. Ridge remains the interpretable reference: its coefficients recover the trend and the market elasticity seen in the exploratory analysis almost exactly.



Holdout residuals for the final candidates: centred, symmetric, and without structure across distance bands or weeks.

6. Fixed-lane December forecast

The chart below is produced by the provided `score.py` from the completed `december_chart_inputs.csv`. The 31 rows fix the lane (Lexington to Fort Wayne, 360 miles), equipment (Dry Van), and weight (32,000 lb), and vary only the date. The model also expects coordinates, a market index, and a quote signal for each row, which are completed from data already available: coordinates from the city reference, the market index from the same-day mean across the December loads in the validation set, and the quote signal from the Dry Van training median. The weekly cycle and the movement of the market index are therefore the only sources of variation in the chart.



Predicted rate for the fixed lane on each day of December 2025.

7. Reproduction

The repository contains the source data, a Poetry environment pinned to Python 3.12, unit tests, and two executed notebooks (exploratory analysis and model selection). The full pipeline is three commands:

`freight-train --models blend --final-model blend`, `freight-predict`, and the provided scorer.

Comparison tables are written to `reports/` by `freight-train`.

Limitations. The linear trend is extrapolated two months ahead; this is appropriate for the horizon but would need a review for longer ones. If the hidden validation labels contain corrupted rates at the training rate, roughly 2.5 points of MAPE are unavoidable for any model.